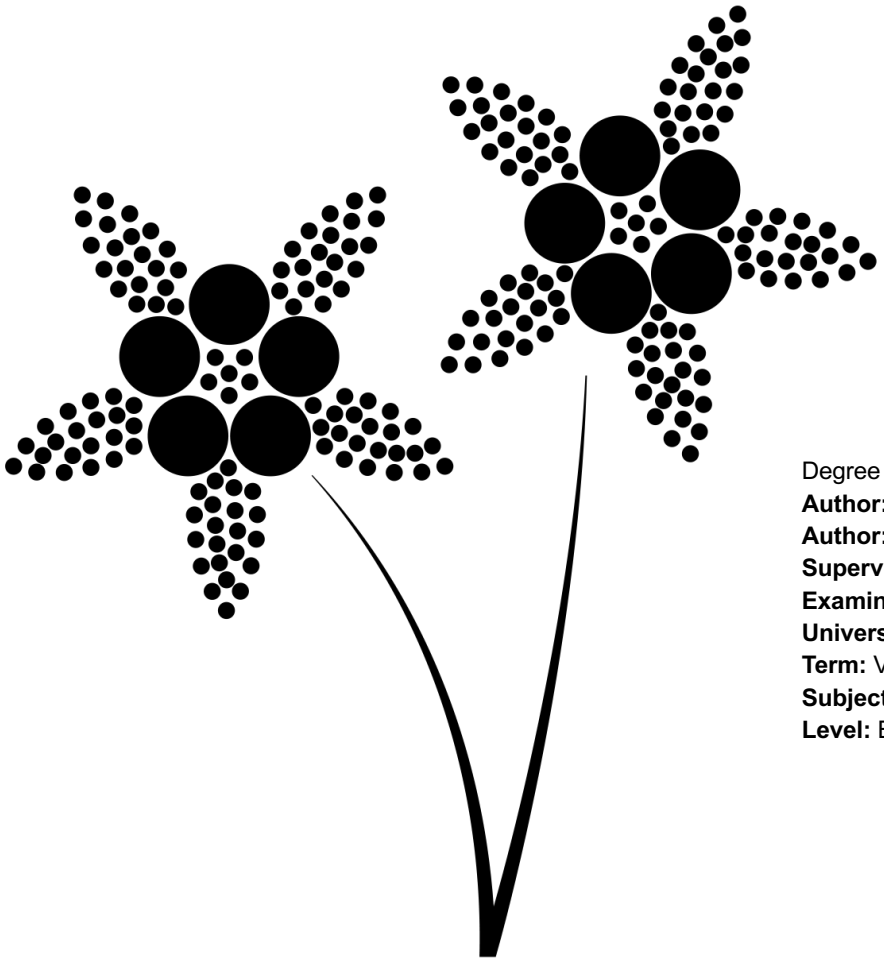
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Behaviour-Influencing Design Elements for Reducing Procrastination Related to Smartphone Use

A comparative study of two mobile applications using A/B testing and interviews



Degree Project
Author: Tea Sjöblom Engvall
Author: Linnea Karlsson
Supervisor: Juan P. Velásquez
Examiner: Päivi Jokela
University: Linnéuniversitetet

Term: VT26
Subject: Informatics
Level: Bachelor

Linnéuniversitetet

Abstract

People tend to increase their time spent using their smartphone which can result in negative consequences like replacing other activities, negative effects on well-being and procrastination. Earlier studies have separately studied how to decrease procrastination and problematic smartphone use with different behavioural-design approaches, but it has been limited to no previous studies targeting the problem of procrastination caused by problematic smartphone use with behaviour-influencing elements.

The study targets procrastination related to smartphone use with behaviour-influencing design comparing two different behavioural-influencing design approaches, nudges and Behaviour Change Techniques (BCT), to see which was most effective in targeting procrastination related to smartphone use. A/B Testing was used to compare the two approaches where two test groups used either a nudge-based intervention or a BCT-based intervention during a test period of seven days. After the test period semi-structured interviews were conducted to collect qualitative data regarding the experience and effects of the behavioural-influencing design elements. Smartphone addiction and screen time was also measured before and after the test period using the Smartphone Addiction Scale – Short version and the participants smartphone statistics over screen time. Lastly, thematic analysis was applied to analyse the results.

Results showed no distinct effect on procrastination, although the nudge-based intervention indicated a somewhat better improvement. Also, the findings indicated that the nudge-based intervention decreased habitual and unintentional smartphone usage, making it more purposeful. Additionally, the BCT-based intervention demanded stronger motivation from the user to get engagement.

Keywords

Procrastination, Behaviour-Influence, Problematic Smartphone Use, Nudges, Behaviour Change Techniques

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1 Introduction

Procrastination is a destructive behaviour pattern that approximately one in five people report struggling with (Kirchner-Krath et al., 2025). The phenomenon can be described as consciously postponing the completion of a task, despite the awareness that potentially negative consequences may follow (Pereira et al., 2025). For individuals who experience this behaviour, it can have significant implications for their quality of life, as procrastination has been linked to increased anxiety, stress, and depression (Kirchner-Krath et al., 2025).

The causes of procrastination are multifaceted and include poor self-regulation, low motivation, and a lack of self-efficacy (Pereira et al., 2025). Although smartphones have simplified everyday life in many ways, overuse has been linked to negative consequences such as procrastination and addictive behaviours (Li et al., 2020). Studies show that the average smartphone user opens their smartphone about 150 times a day (Lukas & Berking, 2018), and that almost half of these users say they think they would have a hard time getting by without it (Li et al., 2020). Previous research that has specifically focused on procrastination in relation to smartphone use has often done so in an academic context, where it has been found that procrastination is linked to poorer academic performance but also to reduced life satisfaction (Xu et al., 2026).

Today, many digital design solutions have been developed to combat procrastination, including mobile applications (Kirchner-Krath et al., 2025). These often use principles from persuasive design and gamification to encourage positive behavioural changes. Previous research also indicates that such design strategies have the potential to increase motivation and reduce procrastination (Pereira et al., 2025).

Despite the increasing number of mobile applications designed to address procrastination, there is still limited understanding of how specific behaviour-influencing design elements contribute to reducing procrastination in everyday smartphone use. Many existing applications combine several strategies, such as gamification, reminders, and feedback mechanisms (Kirchner-Krath et al., 2025; Lukas & Berking, 2018; Pereira et al., 2025), making it difficult to identify which particular design elements influence users' behaviour and experiences. As a result, further research is needed to better understand how these design elements function and how users perceive their effectiveness in supporting behavioural change.

From an interaction design perspective, mobile applications that aim to influence user behaviour represent an important area of study. Interaction design increasingly explores how digital systems shape users' attention, habits, and decision-making in everyday contexts through behaviour change interventions and digital nudges (Michie et al., 2015; Olson et al., 2023). Applications that aim to reduce procrastination often rely on behaviour-influencing design elements such as feedback, rewards, reminders, or gamified

interactions (Kirchner-Krath et al., 2025; Lukas & Berking, 2018; Pereira et al., 2025). Understanding how these elements are experienced by users can contribute to interaction design knowledge about how digital systems can support positive behavioural change.

The problem that this study aims to address is based on the link between problematic smartphone use and procrastination that earlier studies have found (Geng et al., 2018; Li et al., 2020; Olson et al., 2023; Xu et al., 2026). The study aims to investigate this from a design perspective, focusing on how existing digital solutions that use gamification and persuasive design can be used to reduce procrastination in an everyday context. Unlike previous research, which has often focused on examining this from an academic perspective (Li et al., 2020; Xu et al., 2026), the intention is to examine it in a broader context, because there is enough space to explore this in greater depth. Further, whether and why existing design solutions work will be investigated, in order to contribute further knowledge about which design elements appear to be most effective in this context. This topic is relevant due to procrastination and excessive smartphone use are issues that affect many people today (Brockmeier et al., 2025; Geng et al., 2018; Kirchner-Krath et al., 2025; Olson et al., 2023). By investigating which design elements appear to be effective in this regard, it is also possible to contribute to increased knowledge when it comes to the development of future solutions.

Based on this, the following research question has been formulated:

Which behaviour-influencing design elements appear most effective in addressing procrastination related to problematic smartphone use, and why?

1.1 Thesis Structure

This thesis is structured as follows:

- Chapter 2 reviews the literature and establishes the theoretical framework.
- Chapter 3 describes the methodology used in the study and what methods were applied. The study's ethical approach is also presented.
- Chapter 4 presents the results from the data collected relevant to the research question.
- Chapter 5 analyses the empirical material gathered, interprets the findings, defines patterns of themes and connects it with previous research and theoretical frameworks.
- Chapter 6 discusses findings and reflects on the methods. Limitations of the study and contribution to interaction design is also discussed.

- Chapter 7 draws conclusions, discusses societal and ethical implications, and outlines future work.

2 Background and Theoretical Framework

In this chapter, the background and theory for the study will be presented. This includes a literature search, literature review, and theoretical framework. The literature presented in this chapter will be about problematic smartphone use and procrastination, behaviour-influencing design in digital systems, design approaches for reducing procrastination and design approaches for reducing problematic smartphone use.

2.1 Search Strategy and Scope

A literature search was conducted in order to identify relevant studies for the selected field of research, where Scopus was used as the primary database in which the searches took place. OneSearch was used as well, but as a complementary tool in order to verify the effectiveness of the searches and to reassure that no central studies were overlooked.. Throughout the process the results were filtered by English as language and articles as document type. It was also decided that the articles had to be peer-reviewed in order to be included. The initial searches focused on topics of procrastination, behaviour design, smartphone use and persuasive design, which throughout the process resulted in the search string (see Appendix 1. Search String).

The search string resulted in a total of 18 articles in Scopus, which were reviewed in order to determine their relevance for the study. This scope was deemed sufficient as the resulting articles showed potential of being of value as a base for the study due to their subjects. The selected literature was then chosen based on relevance to themes of problematic smartphone use, procrastination, behaviour design, as well as existing design solutions connected to these themes. This resulted in 12 articles selected to be part of the research going forward.

2.2 Problematic Smartphone Use and Procrastination

The amount of smartphone users are increasing globally, and with it the amount of time that people spend on the device (Olson et al., 2023). Over 4 billion people are using social media platforms and their smartphone gives them constant access to always interact and communicate with others (Brockmeier et al., 2025). The usage of the smartphone can be problematic and addictive and cause negative effects such as depression, anxiety and replacing other activities you would rather spend time on (Olson et al., 2023). The increasing usage of smartphones and social media has led to concern about the users' well-being (Brockmeier et al., 2025).

Research shows that smartphone usage is linked to both positive and negative outcomes for users, and that these aspects can be closely linked to one another. Many people use the device in order to communicate and keep in touch with loved ones, while at the same time experiencing addiction and external pressure to be readily available to answer messages and calls (Agarwal & Lu, 2021). The addictive nature of smartphones may leave users

experiencing withdrawal symptoms, compulsive behaviour, disturbed sleep patterns as well as procrastinating tendencies (Li et al., 2020).

One area that has been researched in relation to this is problematic smartphone use and academic procrastination. Li et al. (2020) suggests that academic procrastination rooted in smartphone use may be a maladaptive strategy in order to cope with stressful situations, by putting off the challenging tasks in favor of entertainment. Xu et al. (2026) proposes that elements such as endless scrolling, which is found in apps like TikTok, contribute to immediate reward responses which disrupt the user's focus and ability to self-regulate. This may result in avoidance of performing tasks and working towards a desired goal, which reinforces tendencies towards procrastination and other negative effects. Both these articles point towards a correlation between problematic smartphone use and academic procrastination, with Xu et al. (2026) having an emphasis on TikTok. It is described how prolonged procrastination may lead to consequences such as addiction, cravings, a reduced sense of self-worth, overall reduced satisfaction of life and an increased feeling of loneliness (Li et al., 2020; Xu et al., 2026).

Brockmeier et al (2025) addresses a few motivators that social media users might have for lowering their usage. One of these is that scrolling might feel like a time-wasting activity where time seems to pass by faster than they realize. This time could be spent on activities that would increase the users' well-being, like sleep or face-to-face interaction. Another motivation is that social media use might interrupt daily activities through notifications or messages, which can lead to a feeling of overload and undesired multitasking. A third motivation to decrease social media use is to not get exposed to harmful content, like idealized social media portrayal. This can be a negative influence on the users' well-being since the user feels the need for constant validation.

Another problematic behaviour pattern linked to procrastination is that of internet addiction, described as excessive use that damages individuals' social and psychological functioning, has also been linked to procrastination. To investigate the link between internet addiction and procrastination, Geng et al. (2018) established five hypotheses: (1) Negative core self-evaluation leads to internet addiction, which in turn leads to procrastination, (2) Negative self-control leads to internet addiction, which in turn leads to procrastination, (3) Core self-evaluation and self-control sequentially mediate the effect of procrastination on internet addiction, (4) Social conformity moderates the effect of internet addiction and procrastination, and (5) Social conformity moderates the effect of internet addiction and core self-evaluation (Geng et al., 2018).

In the study, Geng et al. (2018) present that their research has found a positive correlation between internet addiction and procrastination among young adults in China. They were also able to show a positive correlation between core self-development and self-control. They also found that both internet addiction and procrastination correlated negatively with core self-development and self-control, respectively. Social adaptation was also found to correlate positively with core self-development.

2.3 Behaviour-Influencing Design in Digital Systems

To change a behaviour an active ingredient has been proven to be effective when applied in an intervention. Michie et al. (2015) describes these active ingredients as Behavior Change Techniques (BCTs) and lists, describes and structures them in the Behavior Change Techniques Taxonomy (BCTT). The BCTT contains 93 different BCTs clustered in 16 groups (see Table 1). The taxonomy is well developed and 400 participants took part in the various parts of the design process, including international behaviour change experts, researchers, and systematic analysts. The goal was to provide a clear framework for research. With a structured list and definitions of the BCTs it makes it easier for research to develop, and use interventions in a consistent way. The taxonomy intended to promote the development of more effective solutions and to assist with behaviour change techniques that have been proven to work.

Table 1
Clusters of Behavior Change Techniques (BCT)

Group	Title	Description
1	Goal setting and planning	Focus on clear goals and creating structured plans to achieve them. Translate the desired outcome into practical steps.
2	Feedback and monitoring	Focus on tracking the behaviour and giving feedback to improve or adjust, and creating awareness.
3	Social support	Focus on social support that can provide assistance, both emotional and practical, and can help with motivation.
4	Shaping knowledge	Focus on improving skills related to the behaviour.
5	Natural consequences	Focus on emphasising what natural consequences can occur if continuing with current behaviour, can be related to areas like health or environment.
6	Comparison of behaviour	Focus on comparison of people's behaviour to get a reference point. Seeing others' behaviour can make a person understand what is considered the norm and want to follow.
7	Associations	Focus is to link behaviours to different associations like cues. The goal is that the association can trigger to stop an unwanted behaviour, or encourage a wanted one.

8	Repetition and substitution	Focus is to repeat desired behaviours, and to practice implementing substitutes to unwanted behaviours.
9	Comparison of outcomes	Focus on comparing different outcomes of behaviours, to make informed decisions.
10	Reward and threat	Focus is to use reward and punishments to influence a behaviour.
11	Regulation	Focus on learning to regulate internally. This includes factors like emotions, stress, and other psychological states.
12	Antecedents	Focus is to modify external factors such as the surrounding or triggers, to make desired behaviour easier.
13	Identity	Focus on how individuals see themself and their values. The aim is to align a behaviour with a person's identity.
14	Scheduled consequences	Focus on applying rewards or punishments based on behaviour over time to gradually influence the behaviour.
15	Self-belief	Focus is to strengthen the individual's confidence to perform a behaviour.
16	Covert learning	Focus on covert learning and how individuals can learn mentally about a behaviour to influence it.

Note. The 16 categories of behavior change techniques presented in the taxonomy of behavior change techniques (Michie et al., 2015).

Another design approach targeting behaviour change is nudges. Nudges are explained as small changes in the environment with the intent to influence people's behaviour without restricting them. To explain how nudges can be effective, the framework the Fogg Behavior Model can be applied (Olson et al., 2023). Fogg (2009) explains that a behaviour results from a combination of motivation, ability and prompts. Based on Fogg's model the habitual behaviour of problematic smartphone use can be described with a notification being the prompt, the easy access to picking up the smartphone being the ability, and the motivation is scrolling on the smartphone because of boredom. To reduce this behaviour the same theory can be used (Olson et al., 2023).

Olson et al. (2023) present ten nudge strategies (see Table 2) that can help reduce smartphone usage by targeting the users motivation, ability, and prompt.

Table 2
Nudge strategies by Olson et al. (2023)

Number	Title	Description
1	Notifications	Turn off unnecessary notifications.
2	Accessibility	Turn off the sound, put the screen downwards and keep it out of sight.
3	Unlocking	Turn off easy unlocking mechanisms like Face ID and Touch ID, and use password.
4	Sleep	Keep the phone on silent and out of reach.
5	Display	Make it greyscale, turn down the brightness of the display, and change to warm colors.
6	Social Media	Hide apps or delete, make them less accessible.
7	Computers	If you can use your computer instead of your phone, do it.
8	Relationships	Let close friends and family know that you will respond less frequently.
9	Presence	Leave your phone when you don’t need it.
10	General	Use your phone the least possible.

Note. Description of the nudge strategies (Olson et al., 2023).

2.4 Design Approaches for Reducing Procrastination

Procrastination is a common behavioural pattern that affects approximately one in five people (Kirchner-Krath et al., 2025). There has been some research done exploring how design can be used in order to combat this issue. One strategy that has been identified is gamification, which is described as the integration of game-like elements into environments where the game itself is not the ultimate goal, but rather a tool for fostering positive behaviour. When applied correctly it can be a useful tool in encouraging positive behaviour and increasing the motivation in an individual by making tasks more fun and stimulating, as well as decreasing the feeling of overwhelm (Pereira et al., 2025).

According to Pereira et al. (2025) it is important to target the individual's motivation in order to combat procrastination, whereby they identified a framework based on intrinsic and extrinsic motivational factors of gamification. The extrinsic ones include points,

badges/emblems, leaderboards, progress bars, group competitions, mini-games, levels, and visual representation, while the intrinsic ones consist of challenges, avatars, narrative/storytelling, interactions, phase division, resource management, and chatbots/NPCs. They found that a well-crafted design where these motivational factors are used intentionally and balanced may be effective against procrastination. For example, tasks that require greater effort without direct rewards can benefit from including badges or mini-games (extrinsic motivation), but can also be made more stimulating by including elements of storytelling or challenges (intrinsic motivation) (Pereira et al., 2025). One study tested a digital design solution that used persuasive design (PSD) and gamification elements. The design solution provided the user with immediate response by a happy or frowning smiley-face depending on if they answered right or wrong on stimuli connected to procrastination, as well as a text saying “correct” or “that’s wrong”. The user would also earn stars to further encourage the desired behaviour. According to the study, this type of digital solution has the potential to create, encourage, and maintain positive habits in the user as the participants showed a significant reduction in behaviours related to procrastination (Lukas & Berking, 2018).

Kirchner-Krath et al. (2025) studied and identified the main design archetypes used in mobile applications aiming to combat procrastination, based on the Behavior Change Technique Taxonomy (BCTT) and Lowry et al.'s classification of intrinsic motivation for system use, where the first one focused on functional design and the second on hedonic (pleasure and feeling) or eudaimonic (meaning and development) design aspects. This resulted in five main clusters of design archetypes, based on 127 mobile applications aiming for positive behaviour change in individuals who procrastinate. The five design archetypes are (1) Structured Progress Monitoring, (2) Self-improvement guide, (3) Productivity adventure, (4) Emotional coach, and (5) Social focus partner. Cluster 1 was most common in 65 apps, followed by clusters 2 and 5, which were found in 20 apps each.

Cluster 1, Structured Progress Monitoring, is primarily characterized by managing personal goals, tasks, and habits. The user can decide specific time frames for how often and when a task should be performed, and notifications remind the user to complete their list of tasks. The app also presents graphical elements of the user's statistics and visual elements that show the user's progress. These can serve as incentives to maintain the desired behaviour and can be presented as progress meters, grades, badges, points, and streaks (number of days the tasks have been completed without interruption) (Kirchner-Krath et al., 2025).

Cluster 2, Self-Improvement Guide, may include AI chatbots that can help the user with personal advice on changing their behaviour patterns and forming new habits. The user is encouraged to share reflections on their behaviour, which creates opportunities for self-insight but also for personal guidance. To maintain the desired behaviour, design elements such as streaks are used as encouragement and challenges (Kirchner-Krath et al., 2025).

Cluster 3, productivity adventure, has clear elements of hedonic and eudaimonic design. Here, playful content is seen in the form of stories, mini-games, progressive levels, and customizable characters. These types of content can have a positive effect and evoke feelings of enjoyment, challenge, curiosity, and self-expression. The basis for this cluster is that the user defines their tasks step by step, with a timer that schedules sessions for focus, notifications of success, and rewards for completing tasks. The digital character in the app also accompanies the user and provides confirmation and companionship along the way (Kirchner-Krath et al., 2025).

Cluster 4, Emotional Coach, includes content such as quotes, images, videos, and audio to motivate the user but also counteract any perceived pressure. The focus here is on working on the user's self-belief and encouraging a motivating and inspiring view of themselves (Kirchner-Krath et al., 2025).

Cluster 5, is characterized by the main function of a timer that limits the use of other apps as possible external distractions. The user specifies the tasks they want to complete along with a time interval. If the user succeeds in completing their tasks within the time frame, they are rewarded, while if the user fails, they are subjected to a digital punishment. Common hedonic elements in this cluster are fictional characters that develop and behave according to how the user performs. If the user succeeds, the characters are rewarded, while if the user fails, the character is subjected to punishment (Kirchner-Krath et al., 2025).

2.5 Design Approaches for Reducing Problematic Smartphone Use

Previous studies show that problematic smartphone use can have a negative impact on the user. To reduce this problem different design approaches have been studied.

Ertemel and Ari (2020) address problematic smartphone use through a marketing approach. The authors used the concept of the hook strategy, based on Persuasive Technology, where a four step loop is used in order to create habitual behaviours in consumers. The idea is that this loop will increase the likelihood of repetitive behaviour by firstly introducing a trigger (1), which then encourages action (2), rewarding (3) said action and lastly having the user invest in some sort of capacity. In the study they use the hook strategy as well as elements of gamification in order to create an education program where the goal was to “unhook” the participants from the addictive nature of their smartphones (Ertemel & Ari, 2020).

The education program consisted of a PowerPoint presentation and videos, and were divided into eight modules that the participants participated in during two days. The first modules created awareness of problematic smartphone use. They included tools on how to unhook the participants from their smartphones. The following module introduced a digital diet program. The digital diet program proceeded throughout the rest of the

modules including different attention focusing techniques and strategies. The results showed that the program had been successful in decreasing compulsive and unconscious smartphone usage (Ertemel & Ari, 2020).

Both Olson et al. (2023), Liu et al. (2025) and Brockmeier et al. (2025) approach problematic smartphone use with nudge-based interventions. Nudges targeting smartphone use can be described as small adjustments in the user's smartphone settings or environment with the intent to decrease smartphone usage. Olson et al. (2023) Ten nudge strategies that can lower smartphone usage by targeting the users motivation, ability and prompt have been used by both Olson et al. (2023) and Liu et al. (2025) in their separate studies.

Olson et al. (2023) tested the effect of nudges on problematic smartphone use. The participants got introduced to the ten nudges and decided on which nudges to implement and to what degree. The results from the study showed that the implementation of the nudges reduced problematic smartphone use, screen time, and improved sleep. The time that no longer was spent on the smartphone was among several participants instead used for studying, hobbies or socialising (Olson et al., 2023).

In Liu et al. (2025) study the problem of smartphone addiction (SA) was addressed in association with bedtime procrastination (BP), and chronotype (the individuals preferred sleep time, early or late). One of the studies included in the research where the aimed to lower SA with Olson et al. (2023) ten nudge-based strategies, and as a result of that also lower BP while assessing the users chronotype (Liu et al., 2025).

Results from the study presented lower levels of SA, screen time, and more morningness among the participants in the intervention group. Nudge-based intervention targeting SA can help reduce consequences like BP (Liu et al., 2025).

In Brockmeier et al. (2025) research an intervention app with customizable nudges was tested to see what effect it had on the users self-regulated social media use, and what positive and negative effects that it can have on the users well-being and stress levels. They used a randomized controlled trial with a total of 70 participants. 35 participants tested the intervention with access to its functions and 35 participants concluded the control group. The user got to customize their nudges by choosing which social media apps they felt were problematic. They also got to choose their own goal for how much time they wanted to spend on the app, and how often. The users picked the nudges they preferred to receive and how often. Lastly, the users also described activities they wanted to perform instead of spending time on social media.

The result of Brockmeier et al. (2025) study showed that there was no significant difference between the intervention group and the control group regarding well-being, positive effects, negative effects or stress. Though, some differences between the groups was found in the last questionnaire that was conducted post-usage of the intervention app.

The intervention group reported higher well-being, less negative effects and lower stress levels. In the result it was reported that users with higher levels of problematic smartphone use experienced lower levels of stress among the participants in the intervention group.

2.6 Theoretical positioning

Based on the aforementioned studies and the insight that procrastination can be a consequence of smartphone addiction (Li et al., 2020; Xu et al., 2026; Geng et al., 2018), as well as previous research on mobile applications with behaviour-influencing designs to address procrastination (Kirchner-Krath et al., 2025; Lukas & Berking, 2018; Pereira et al., 2025), and the potential of design approaches to reduce smartphone usage (Brockmeier et al., 2025; Ertemel & Ari, 2020; Liu et al., 2025; Olson et al., 2023), this study aims to investigate how the behavioural-influencing approaches BCTs and Nudges, compares when targeting procrastination in problematic smartphone use.

There is currently limited research on behaviour-influencing design in relation to procrastination, and to our knowledge, no research specifically investigating behaviour-influencing design targeting procrastination caused by problematic smartphone use. Given the established connection, this study is considered highly relevant.

This study will use Michie et als. (2015) Behavior Change Technique Taxonomy as a framework for describing which BCTs are being incorporated and assessed in the study when specifically targeting procrastination. To target behaviour connected to smartphone usage Nudges and Olson et al. (2023) ten nudge strategies will be included as a framework. Finally, to understand a behaviour and how to change it The Fogg Behavior Model serves as a foundation.

3 Methodology

In this chapter the methodology of the study will be presented. It will present what methods were used to collect data and analyze the result, the target group, and the studies ethical approach.

3.1 Research Design and Framing

Through a literature review, the current state of research about mobile applications with behaviour-influencing designs aimed at combating procrastination (Kirchner-Krath et al., 2025; Lukas & Berking, 2018; Pereira et al., 2025), and procrastination as a consequence of problematic smartphone use (Li et al., 2020; Xu et al., 2026; Geng et al., 2018) was explored. Based on this, the research question was framed, where the focus of the study became attempting to identify which behaviour-influencing design elements appear to be effective against procrastination caused by problematic smartphone use.

The research paradigm of this study is interpretivism, or constructivism. To conduct the study, a quasi-experimental comparative approach was selected. The selected methods were A/B testing and semi-structured interviews. This was deemed fitting due to the nature of the study, where the effect of two different applications were compared to one another. Through the A/B testing the Smartphone Addiction Scale – Short Version (SAS-SV) together with statistics of the users screen time, was used as a complementary quantitative measure, in order to further explore the effect of the design elements of the applications. Since the research question for this study was “Which behaviour-influencing design elements appear most effective in addressing procrastination related to problematic smartphone use, and why?” This approach was considered because of the use of methods that allowed for comparing different design elements. It was also concluded that a qualitative approach, through the interviews, would be of interest in order to understand participants’ lived experience.

3.2 Selection of Applications and Design Elements

This study compared two different mobile applications with different implemented behavioural-influencing design approaches, one targeting procrastination and the other targeting smartphone usage.

Application 1 targeted procrastination with BCTs, complemented by gamification elements. This app was chosen because it included distinct BCTs specifically targeting the behaviour of procrastination (Kirchner-Krath et al., 2025). Also, the app included gamification elements that in previous research have been shown potentially useful against procrastination (Lukas & Berking, 2018).

The application was included in Kirchner-Krath et al. (2025) study as an application targeting procrastination. The app was categorized into the cluster Social Focus Companion and included BCTs from six different BCTs groups (1) Goal setting and

planning, (2) Feedback and monitoring, (6) Comparison of behaviour, (10) Reward and threat, (12) Antecedents, and (14) Scheduled consequences. The function of the application is to initiate a timer when you want to focus on a specific task (see Figure 1). While the timer is going you cannot access other applications on your smartphone. When the time is up the user will get a reward, a virtual plant contributing to a growing digital garden.

Figure 1
Application 1

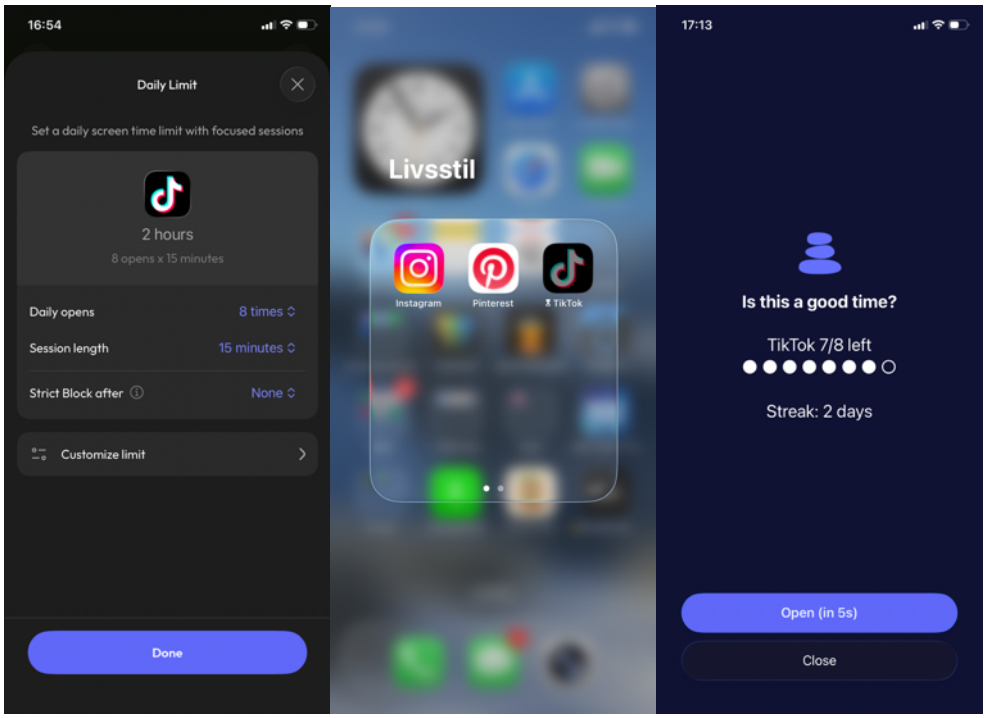


Note. Screen pictures from Application 1. (Left) Showing the timer function the users start when they intend to focus. (Middle) Presenting how the application on the user's home screen gets restricted while the timer is on. (Right) Pictures how the garden the user can be grown and built when successfully focused for the whole timer-session.

Application 2 aimed to target problematic smartphone use. It included three of Olson et al. (2023) ten nudge strategies (see Figure 2). The nudges included in the application were (1) Notifications, where the user gets notifications when they open a restricted application or when their preferred time is up for each session using the app, notifications from the restricted application also get restricted (2) Accessibility, the restricted application gets harder to use when restricted due to looking unavailable, the user have to actively choose to confirm they want to open the application, and when they want to open the application it takes five seconds, and (6) Social Media, the applications allows users to restrict all applications that are distracting, including social media applications. Some nudges could be customised by the user according to each individual's smartphone use. Each participant could pick and limit applications they particularly thought were

distracting and restrict it according to their own preferred goal. Then, the application nudged the user with preferred fullscreen prompts and notifications to use their smartphone less. This app was chosen because previous studies have pointed out positive results from nudge-based interventions against smartphone usage (Brockmeier et al., 2025; Liu et al., 2025; Olson et al., 2023).

Figure 2.
Application 2



Note. Screen pictures of Application 2. (Left) Presenting the user choosing how many times they were allowed to open their chosen application and how long each session should be. (Middle) Presents how the restricted application looks on the user's homescreen. (Right) Shows how the notification looks when the user clicks on the restricted app, or when the session for the app is up.

3.3 Participants and sampling

Participants were recruited using a convenience sampling method (Jacobsen, 2017). Potential participants were approached in informal, everyday contexts and asked whether they wanted to participate, with the recruitment focusing on individuals who reported that they experience procrastination associated with perceived problematic smartphone use.

The inclusion criteria for the participants were that the individuals had to be an adult, meaning at least 18 years old, and experience procrastination that they associated with their smartphone use in a way that affected their everyday life. No restrictions were placed on gender, occupation or other demographic variables, as the study aimed to explore the phenomenon from a general and everyday context. The age requirement was

put into place due to ethical aspects, to make sure that the participants were eligible to give informed consent (Jacobsen, 2017).

Problematic smartphone use was not defined according to a standardized measure/criteria prior to recruitment, but was based on the participants' acknowledged limitation explicitly linked to their smartphone use. If they experienced procrastination linked to what they experienced as problematic smartphone use they were deemed as eligible to participate in the study.

A sample size of a total of 12 participants was considered appropriate given the qualitative and explorative nature of the study (Jacobsen, 2017).

3.4 The intervention

Each participant was individually introduced to the study by one of the two authors. First, an informed consent (see Appendix 2. Informed Consent Form) was read and signed by the participant to ensure that all ethical measures were fulfilled. To see how the participants' procrastination, smartphone addiction and screen time changed during the test period, quantitative measures were collected both before and after the test. The participants got to answer four questions (see Appendix 3. Questions before the test period) about their own experiences of problematic smartphone use, procrastination, and which tasks they tend to postpone in their daily life, including one control question to ensure the participant met the inclusion criteria. These questions were included to understand each participant's assessed procrastination in relation to their smartphone use. Then, the participants answered an online form consisting of the Smartphone Addiction Scale - Short Version (SAS-SV) to understand the participants' assessed level of smartphone addiction, and shared the weekly overview of their smartphone statistics over screen time the previous week. The SAS-SV and statistics of the screen time were collected both before and after the A/B test to compare how these measures changed for the participants in the testgroups. SAS-SV is a measurement to assess people's smartphone addiction. It consists of 10 items about the users' assessed smartphone usage (see Appendix 4. SAS-SV). The participants answer the questions on a 6-point Likert scale between 1 (highly disagree) and 6 (highly agree). Participants then get a score between 10 and 60, where scores above 31 in men and 33 in women is considered as smartphone addiction (Kwon et al., 2013).

To compare the two different behaviour-design approaches, nudges against BCTs and gamification, A/B testing was conducted. A/B testing is a comparative method where two versions of a software are tested. The test focuses on the end-users' point of view to help make decisions in software development (Quin et al., 2024). In this study two different applications were tested against each other. All the participants were divided into two equally sized groups. Each test group got assigned an application, Test Group 1 was assigned Application 1, and Test Group 2 was assigned Application 2. All participants got an introduction to their assigned application where the authors made sure the onboarding

was standardised for everyone, and key functions and the task were clearly presented. Each test group used their assigned application for a total of seven days, allowing participants to engage with the intervention across both weekdays and the weekend.

After the test period semi-structured interviews were conducted with each participant. The purpose of the interviews was to complement the quantitative measures and to collect qualitative data about the participants' experience using the app to get a deeper understanding about the effects of the implemented behavioural design elements. All the interviews were audio recorded to ensure reliability and accuracy.

3.5 Analysis

3.5.1 Thematic analysis

In order to analyse the qualitative data collected through the interviews (qualitative) and the A/B testing results (quantitative) a thematic analysis was conducted (Braun & Clarke, 2006). Since there were two separate test groups involved, where each tested a different application, they were initially analysed separately with thematic analysis. This was done in order to understand and evaluate the results from both applications, as to gain an understanding of how effective the included design elements appeared. The results from the two test groups were then compared to each other in order to evaluate the different effects of the design elements.

The thematic analysis followed the six-phase analysis framework that Braun & Clarke (2006) describes. This means that the data was firstly read through and processed thoroughly, followed by initial coding in order to capture meaningful data segments. Themes were identified through the codes. These were then reviewed and refined until the themes were finalised. The themes were then defined and named, constructing an analytical narrative, and lastly compiling and linking the analysis back to the research question and literature (Braun & Clarke, 2006).

3.5.2 SAS-SV and screen time

The data that was collected from the SAS-SV prior to the study was compared to the results after the study. The results were analysed in order to explore if the participants' scores had changed throughout the duration of the study. Similarly, screen time data. The data collected from before the study was compared to the data from after the study in order to see if the screen time had been affected by the application that was tested out.

3.6 Ethical Approach

Throughout the study, ethical aspects were taken into consideration. The study was considered ethically defensible due to it not causing any physical or psychological harm to those involved in it (Jacobsen, 2017). Before agreeing to participate in the study informed consent was obtained from all of the participants (see Appendix 2). Potential participants were presented with clear instructions of the purpose of the study, what

participation entails, and how sensitive data will be handled, before giving their written consent to participate (Nyberg & Tidström, 2012).

Jacobsen (2017) specifies four main components of informed consent: Competency, voluntary participation, complete information and comprehension. In order to make sure that the participants could give their informed consent, a criterion was established that the participant had to be an adult (18 years or older). Through the informed consent form, as well as oral instructions, potential participants were also informed that participation in the study was completely voluntary and that they could withdraw their consent at any time, and that their data would not be used in that case. The aim was to present all relevant information in a transparent way, as well as create an open dialogue with the participant.

The nature of some of the collected data might be considered as more sensitive, such as screen time and the results from the SAS-SV, since it can indicate what could be considered as bad habits and tendencies of addiction-like behaviours. Given this, it was decided that the participants would be reported confidentially, with no identifying factors being revealed about them throughout the study. This includes age and gender, which are not central to this study and that, due to the smaller sampling size, could be revealing of the individual (Jacobsen, 2017). In the consent form, it is stated that the collected data exclusively will be available to the students conducting this study, the supervisor and the examiner, and that it will only be used in order to finish this project. It is also stated that it will be deleted once the study has been completed and approved.

Great care was taken to ensure that the study meets the requirements for individual protection. Measures were taken in order to ensure that the participants understood the implications of participation, and that informed consent was practiced. The data was ensured to be handled securely and without harming any individual. Finally, it was ensured that the data was only used for the purpose of answering the research question (Johansson, 2018).

4 Result

This chapter presents the data collected and descriptive results of the study based on A/B testing, interviews, SAS-SV, and statistics over screen time data. Findings and data from each test group are presented individually, followed by a comparison between the two groups.

4.1 A/B Testing and Interview

A total of 12 participants completed the A/B testing procedure. Participants were recruited to participate in the study, six in each test group. However, one participant from Test Group 1 withdrew their consent after the data collection phase, leaving five participants in this group. According to the consent form, they had the right to do so, meaning that their results are excluded from this study.

Test Group 1 tested Application 1 with BCTs, and Test Group 2 tested Application 2, which included nudges. Individually in the test groups, each participant used the assigned application for a total of seven days. After the test period all participants answered 12 questions (see Appendix 5. Interview questions) in an individual semi-structured interview regarding their experience of smartphone usage and procrastination during the test period.

4.1.1 Test Group 1

Test Group 1 consisted of five participants that each got to try out Application 1 for one week. The interviews were conducted after this period in a setting chosen by the participants.

The initial interview session of Test Group 1 showed that all but one participant experienced that they postponed tasks in their everyday life due to smartphone usage, while the last participant reported not allocating time to these tasks rather than postponing them. All of the participants said that they used their smartphone more than they preferred before the test period began.

The interview that took place after the test period showed that two participants reported that their tendencies towards procrastination had decreased, one participant was uncertain whether their behaviour had changed, and two participants did not notice a difference. Two participants reported a perceived increase in productivity through using Application 1, while three participants experienced that they had not become more productive. One participant said that they did not think that they could tell in this short amount of time.

Three participants reported continued excessive smartphone use after the test period, while two participants reported slightly excessive use. Two of these thought that their smartphone use had decreased.

Three participants liked the timer-function and the fact that they could not use certain applications while it was running. However, it was noted that not every application on the participants' smartphones was blocked, which they were disappointed by. The feature which allowed for the user to allow certain apps was appreciated.

Two participants thought that it was hard to remember to use the application, and one participant did not like the notification feature as it led them to pick up and use the smartphone. Two participants commented that they appreciated the "app mascot", the plant with a happy face.

Three participants noted that the test period had led to indirect effects of their smartphone use that was not directly linked to Application 1. They described that the knowledge that the application was downloaded reminded them to not use their smartphone as much, that their mindset going into the week was to reduce their usage and that they have made the active decision of not always bringing it with them.

Through thematic analysis, three themes and thirteen sub-themes (see Appendix 6. Thematic Analysis Themes) were identified based on 123 codes.

4.1.2 Test Group 2

Test Group 2 consisted of six participants that used Applications 2 for a test period of seven days. After the test period interviews were held with each participant. Using thematic analysis, 189 codes were identified from the interviews with Test Group 2, which was organized into three themes and fourteen sub-themes (see Appendix 6. Thematic Analysis Themes)..

The initial interview prior to the test period, showed that all six participants used their smartphone more than they would like. Five out of six said they were prone to procrastinate tasks in their daily life.

After the test period five out six participants in Test Group 2 reported reduced compulsive smartphone use during the test period. All participants had a good experience with using Application 2, and felt they thought more carefully and reflected more on their smartphone use during the test period. Among other things, they explained it led to more meaningful screen time that served a purpose. All participants experienced a decrease in their smartphone usage, particularly the frequency of smartphone use during the day.

The notifications participants received when they open a restricted application, led them to reconsider if it was necessary or timely to use their smartphone. Three participants reported that they shifted their usage to other apps that were not restricted. For example, one said they played a different game than the one that was restricted.

Four participants said they appreciated the notification they got when their time for the session was up, three reported that it facilitated disengagement.

Three participants perceived the application limitations are restrictive, because they were unable to access information directly, due to the five seconds delay when opening up the restricted application. Restriction also appeared when their usage was interrupted, due to the time limit when they were checking things that served a purpose, such as news.

Application 2 limited notifications of the chosen restricted applications, produced both facilitating and frustrating effects. Three felt that limited notifications reduced unnecessary use of the apps, one felt it was frustrating when important information they wanted did not reach them, and two missed notifications on messaging or communication apps, which led to messages being missed or, conversely, increased usage causing them to open the app more often than necessary.

Three participants said they were not as distracted by their smartphones, which in turn led to better focus. Three felt that they procrastinated a little less during the test period. Three felt that their productivity at work or in their studies had improved slightly during the test period.

4.1.3 Comparative results of Test Group 1 and 2

A comparison between the two applications and test groups, where Test Group 1 had five participants and Test Group 2 had six participants. Both groups had all but one participant say they tend to postpone tasks in their daily life before the test period, and after the test period two participants from Test Group 1 and three from Test Group 2 reported increased productivity. Test Group 1 had two participants experience a decrease in procrastination during the test period while one additional participant in Test Group 2 reported that, which made a total of three.

Two participants in Test Group 1 felt they had decreased their smartphone usage during the test period, at the same time three participants still said they used their smartphone more than they would prefer. In Test Group 2 all six participants experienced decreased smartphone usage during the test period, especially the amount of times they used their smartphone. Compared to Test Group 1, only two participants experienced decreased smartphone use. Three participants in Test Group 1 still thought they had used their smartphone more than preferred during the test period, while Test Group 2 had one additional participant, a total of four experiencing this.

4.2 Screen time

4.2.1 Test Group 1

The results for Test Group 1 showed that three participants reduced their average daily screen time, while one participant increased theirs. One of the participants, Participant G1-5, was unable to share the screen time.

Participants G1-1, G1-2 and G1-3 reduced their daily average smartphone use between 35 minutes and 1 hour 25 minutes, while Participant G1-4 increased their usage by 34 minutes (see Table 3).

Table 3
Screen times Test Group 1

Test person	Daily average		Difference
	Before test period	After test period	
G1-1	4 h	3 h 25 min	– 35 min
G1-2	8 h 4 min	6 h 57 min	– 1 h 7 min
G1-3	3 h 25 min	2 h	– 1 h 25 min
G1-4	6 h	6 h 34 min	+ 34 min
G1-5	—	—	—

Note. Screen times for four of five participant in test group 1, before and after the test period.

4.2.2 Test Group 2

In Test Group 2, results showed that four participants reduced their daily average screen time by between 14 minutes and 1 hour 28 minutes, during the test period compared to before. Two participants increased their screen time, G2-4 by 25 minutes and G2-5 by 26 minutes (see Table 4).

All the participants in Test Group 2 chose to limit some sort of social media applications when they were encouraged to pick the applications they would like to limit while setting up Application 2 during the introduction of the experiment. During the test period results show that all participants decreased the screen time on the application they chose to limit with between 43 minutes and up to 22 hours 46 minutes (see Table 5.).

Table 4
Screen times Test Group 2

Test person	Daily average		Difference
	Before test period	After test period	
G2-1	4 h 32 min	4 h 47 min	+ 15 min
G2-2	3 h 30 min	3 h 16 min	– 14 min
G2-3	2 h 35 min	2 h 15 min	– 20 min
G2-4	7 h 1 min	7 h 26 min	+ 25 min
G2-5	3 h 34 min	4 h	+ 26 min
G2-6	5 h 46 min	4 h 18 min	– 1 h 28 min

Note. Screen times for every participant in test group 2, before and after the test period.

Table 5
Screen time differences in limited app using application 2

		Weekly total		
Test person	Application	Before test	After test	Difference
G2-1	Instagram	9 h 4 min	4 h 27 min	– 4 h 37 min
	Facebook	3 h 24 min	46 min	– 2 h 38 min
G2-2	Instagram	4 h 50 min	4 h 7 min	– 43 min
G2-3	Instagram	3 h 40 min	2 h 13 min	– 1 h 27 min
	Snapchat	2 h 16 min	1 h 33 min	– 43 min
G2-4	Youtube	35 h 44 min	12 h 58 min	– 22 h 46 min
	Facebook	4 h 4 min	3 h 15 min	– 49 min
G2-5	Instagram	2 h 39 min	1 h 37 min	– 1 h 2 min
G2-6	Social applications (total)	24 h 22 min	12 h 40 min	– 11 h 42 min

Note. Screen times in limited application of every participant in test group 2, before and after the test period. The table also shows the difference in time.

Figure 3

Screenprint of smartphone screen time statistics



Note. Screen print from a participant in Test Group 2. (Left) Showing statistics from a week before the test period and the most used app in hierarchical order. (Right) Showing the statistics of the week of the test period.

4.2.3 Comparative results of Test Group 1 and 2

The results from the participants' screen time from before and after the test period showed that both Test Group 1 had three participants lower their smartphone use, while Test Group 2 had four. In Test Group 1 the reduced screen time fell between 35 minutes to 1 hour 25 minutes, while the reduced screen time for Test Group 2 ranged between 14 minutes to 1 hour 28 minutes.

One participant from Test Group 1 increased their screen time with 34 minutes, while one participant was unable to disclose their screen time. Test Group 2 had two participants that increased their screen time by 25 and 26 minutes respectively.

4.3 SAS-SV

Both test groups, all twelve participants, completed the 10-item SAS-SV questionnaire, before and after the test period. A total of 24 completed responses were collected, out of which 22 were used in the study. The average time answering the forms was 3 minutes and 56 seconds.

4.3.1 Test Group 1

Results showed that four out of five participants in Test Group 1 reduced their score after the test period. Before the test period one participant scored 33 points or higher, the threshold indicating potential smartphone addiction. After the test period two participants increased their score by 3 points compared to their prior test but only one participant scored higher than 33 (39).

Table 6
SAS-SV scores for Test Group 1

Test person	Question										Points	Change Before/after
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**		
G1-1 före	2	3	4	1	5	5	3	1	4	2	30	
G1-1 efter	2	5	1	4	4	1	4	1	6	1	29	– 1
G1-2 före	1	1	4	4	3	3	4	1	5	1	27	
G1-2 efter	3	1	1	5	4	4	5	1	5	1	30	+ 3
G1-3 före	1	2	1	6	4	3	2	1	5	3	28	
G1-3 efter	1	1	1	3	3	2	2	2	3	2	20	– 8
G1-4 före	4	4	3	1	3	3	5	4	5	4	36	
G1-4 efter	6	3	1	3	4	4	2	5	6	5	39	+ 3
G1-5 före	2	1	3	1	3	1	3	5	6	1	26	
G1- 5 efter	4	2	4	1	2	1	3	1	6	1	25	– 1

Note. SAS-SV scores for every participant in Test Group 1, before and after the test period.

4.3.2 Test Group 2

The results from the SAS-SV of Test Group 2 concluded that five out of six participants reduced their score after the test period compared to before. Three participants reduced their score by more than 15 points. Before the test period four participants scored 33 points or higher, scores above the threshold for smartphone addiction. After the test period all participants scored below the threshold for smartphone addiction (see Table 7).

Table 7
SAS-SV scores for Test Group 2

Test person	Question											Change Before/after
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**Points**	
G2-1 före	4	4	4	4	2	2	3	2	5	3	33	
G2-1 efter	2	3	2	3	3	2	2	2	4	4	27	– 16
G2-2 före	2	4	4	5	3	3	3	2	5	4	35	
G2-2 efter	2	3	2	5	3	3	4	2	4	4	32	– 3
G2-3 före	3	4	2	6	4	4	5	4	5	4	41	
G2-3 efter	2	4	1	3	1	3	3	2	3	2	24	– 17
G2-4 före	4	4	2	4	4	2	3	5	5	5	38	
G2-4 efter	2	2	2	3	2	2	2	1	2	2	20	– 18
G2-5 före	1	1	3	7	3	1	5	1	4	4	30	
G2-5 Efter	1	1	1	6	3	4	5	2	4	5	32	+ 2
G2-6 före	1	1	1	6	4	2	4	2	6	2	29	
G2-6 efter	1	1	1	5	3	3	5	2	5	2	28	– 1

Note. SAS-SV scores for every participant in Test Group 2, before and after the test period.

4.3.3 Comparative results of Test Group 1 and 2

Comparing the results of the SAS-SV before the test period, Test Group 2 had a higher score (34.3) than Test Group 1 (29.4). After the test period, Test Group 2 lowered their average score by 7.1 points and scored a total average of 27.2, which was lower than Test Group 1's 28.6.

5 Analysis

This chapter includes the analysis of the results. The patterns of the themes from the thematic analysis of the two testgroups are discussed in relation to previous studies and relevant theories. Following, the quantitative data is discussed and interpreted. And lastly, a summary is included to compare the two applications in order to address the research question.

5.1 Analysis of Qualitative data

5.1.1 Test Group 1

5.1.1.1 G1: Smartphone usage

As all of the participants from Test Group 1 perceived their smartphone use as excessive, as well as considered themselves to engage in some form of procrastinating behaviour because of it (one participant explained it as that they did not necessarily procrastinate, but rather did not prioritise these tasks they wished to complete), it is clear that this behaviour is affecting their everyday life. The results of the interviews also clearly suggested that the participants spent time on their smartphones rather than doing tasks that they would rather want to do, such as spending time on hobbies, reading, socializing, exercising, spending time in nature or simply resting. Instead they would use their smartphone for consuming news, playing games, engaging in social media, watching videos and reading. This suggests a pattern of behaviour where the usage of the device interferes with the everyday life of the individual, depriving both physical wellbeing (exercise, spending time in nature and resting), as well as activities that encourage personal growth and values (hobbies, reading and socialising).

However, a couple of the participants did not perceive their smartphone use as purely problematic, but as something that could also be considered as helpful. For example while trying to tackle mundane tasks, where listening to or watching something was suggested as a helpful tool in order to make the tasks less daunting. Smartphones were also brought up as a tool to read and learn something new, or used out of necessity. This tension could be tied to the findings by Agarwal & Lu (2021), where it is reported that a lot of smartphone users experience a balance of problematic smartphone use and healthy habits. This further suggests that the subject of smartphones and the problems they bring is a complex one, where the device in and of itself might not be the primary issue. From a design and behaviour-influencing perspective might not be to completely make the users stop using their smartphones, but foster and encourage a healthy and balanced use.

5.1.1.2 G1: Application experience

Three of the participants were positively inclined towards the timer function of Application 1 (BCT 1 and BCT 12). However, not all of the applications were blocked, such as the web browser, smartphone and messages. According to one participant, this resulted in the application not being as effective as it could have been, as browsing on the

web was one of their problem areas. In this case, the participant had to find the motivation by themself to not use the browser while the timer was activated. As motivation is one of the main problems to target in order to address procrastination (Pereira et al., 2025), this might reduce the overall effectiveness of Application 1. That is, in the case that the user does not possess the motivation and self control to avoid the features that are not blocked, which also partly defeats the purpose of the application. It was also brought up that it did not come naturally to them to start the application in order to start the timer, but that it could be easy to do other things on their smartphone. This is also a problematic aspect, as the goal of the application is to deflect the user from aimless and problematic smartphone use, rather than encourage them to spend time on it.

According to two participants, one of the flaws of the application was that in order for the application to be useful, they would have to be motivated by its features in the first place. One of them expressed it the following way: *"If you are like not super dedicated to trying to earn points then it is like, who cares."* This indicated that the penalty feature in Application 1 (BCT 10 and BCT 14) might not have been enough motivation to use the application the way it was intended to. In the case that the user is not motivated by collecting and earning points in the application, they will most likely not be motivated by the possibility of losing said points by not letting the set timer run its course. Pereira et al. (2025). A couple of the participants also had a hard time remembering to use the application, despite having the notifications turned on. This might suggest that the notifications were not enough motivation for the participants to use Application 1 either. Interestingly enough, they were even brought up as a problem by one participant who experienced them as a prompt to lose focus by picking up their smartphone and thus being sucked into it.

All in all, the participants did not make a lot of comments about the gamification aspects of the application, such as the reward system, streaks and garden-building. However, two participants commented on the "mascot" of the application, describing it as cute and motivating (BCT 2). One participant even described this feature as the best thing about the application, while the other appreciated seeing it on their home screen, making them remember to not use their smartphone as much. This functioned as an indirect motivator for the user that was not directly linked to the functions of the application.

Two of the five participants reported potential interest in continuing their usage of Application 1 after the test period was over, while the rest did not. Both of these categories of participants had reported both positive and negative experiences towards the application and their experience with it. However, it is telling that a majority of the participants did not seem to find it effective or useful enough to continue using it, even though they still experience excessive smartphone use. It does seem as though the features of the application were not motivating enough in order to encourage continual use, which would be very important in order for the application to produce real and lasting results in the everyday lives of the participants.

5.1.1.3 G1: Effects of usage

The majority of the participants reported positive effects after using Application 1. They named effects such as increased self control and focus, as well as a greater sense of relaxation while the application was in use. A few participants also pointed out that just the fact that they had the application visible on their homescreen and knew that it was downloaded on their device made them more conscious of their smartphone use, even when not directly using it. Effects such as more time for other things were reported as well, which is positive, since that aligns with the purpose of the application and the BCTs used in it. Instead, the participants partook in activities such as reading, relaxing and socializing. This is positive from the standpoint that the application had the effect of allowing them to partake in activities that they previously stated that they wanted to do, suggesting that it had a positive impact in prioritizing activities. However, only two of the participants experienced a reduction of procrastination, and two participants reported that they thought their screen time had lowered during the test period. This suggests that the application did not necessarily address the problem of procrastination connected to problematic smartphone use in an effective way. This might have to do with the previously mentioned lack of experienced motivating factors related to BCT 10 and BCT 14, where the threat of punishment in the game-features might not motivate the user to reach their goals of creating better habits.

Similarly, a majority of the participants did not perceive their productivity to have improved. One of these stated that they did not think that they could tell if they had become more productive in this short a time, which is an interesting input from a BCT-perspective. BCTs are supposed to sustainably change behaviours (Michie et al., 2015), meaning that the results might have looked different had the test period been longer. However, as the participants seemingly had a hard time finding the motivation to use the application from the start, there are no guarantees that they would continue using it had the time horizon been longer. This uncertainty is further deepened by the fact that only two of the participants expressed a potential desire to continue using the application after the test period.

5.1.2 Test Group 2

5.1.2.1 G2: Smartphone usage

Test Group 2 declared that all participants used their smartphone more than they would like before the test period. This can be interpreted as the smartphone usage being problematic which supports earlier studies that smartphone usage has addictive tendencies (Li et al., 2020). When explaining what the participants spent time on while using their smartphone, the majority usually use different social media applications, and also online shopping. Playing games, communicating, getting information, news, and researching were also common activities among the participants. Even though the participants thought they overused their smartphone, all time spent on it is not negative, just as explained by Agarwal & Lu. (2021). Communicating with loved ones and getting news and researching can be in the user's interest and be valuable to the participants,

while doomscrolling might make the participants lose track of time unintentionally and lead to negative consequences such as postponing sleep, decreased focus, and replacing other activities. These findings support Brockmeier et al. (2025), and Olson et al. (2023) on problematic smartphone use replacing activities. Also, five out of six participants tend to procrastinate tasks in their daily life, such as chores in the home or work related tasks, which supports the theory of smartphone usage leading to procrastination tendencies (Brockmeier et al., 2025). Adding to this, the participants would rather spend time reading, prioritize health, and doing chores than spending it on their smartphone. This indicates clearly that smartphone usage has negative impacts when replacing other activities, especially once that will increase health and well-being (Brockmeier et al., 2025).

5.1.2.2 G2: Application experience

All participants said using Application 2 was a good experience that made them reflect more on the timing, and purpose, leading to more purposeful smartphone use. This pattern can be connected to the notifications the participants got when opening up a restricted application, making it both harder to enter the application and giving them a choice to reflect over. When connecting this to Olson et al. (2023) nudge strategies, notification and accessibility, it becomes clear that these elements are effective to decrease unnecessary smartphone use. The notification the participants got when their session was also perceived as effective by three participants, while they claimed they often put their smartphone away when the time was up. The nudges again made the participants reflect over their use and gave them an active choice to continue or disengage, and in this case made them choose option two (Olson et al., 2023). Connecting these findings to Fogg (2009) and his Behaviour Model, the notification (nudges) made the participants get a trigger to end or question the behaviour they are currently performing, or aiming to perform, this made them reflect over their motivation to open or reopen the application, and also making the application harder to use, by restricting it, which limits the availability of the application. This indicates that the nudges are effective in targeting all three elements of a behaviour, prompt, motivation and availability, making it possible to change unnecessary smartphone use into a more purposeful one (Fogg, 2009).

5.1.2.3 G2: Effects of usage

The effect shown while and after using Application 2 was that almost all participants felt they did not get stuck on their smartphones, such as doomscrolling. All participants also declared that they had been reflecting over their smartphone use more, leading to less and more meaningful use of their smartphone compared to before. This could be explained as a switch in motivation for using the smartphone, which is one component for explaining a behaviour (Fogg, 2009).

Three participants assessed they used their smartphone less during the test period, which is positive. This could be related to three participants feeling they didn’t get as disturbed by their smartphone during the test period which led to better focus. Getting better focus

could also explain why three participants said they got more productive at work or at studying, and three participants experiencing less procrastination during the test period, compared to before. Limiting notification on restricted application, one of the nudges used in the intervention, can explain this outcome. By getting less notification, the user does not get a prompt, one of the components of a behaviour, which leads to taking up their smartphone, and the user can stay focused on the task they are doing or planning to do (Fogg, 2009).

5.2 Analysis of Quantitative data

5.2.1 G1: Quantitative data

Before the test period one of the participants of Test Group 1 scored 33 or higher on the SAS-SV test. This means that according to this test, one of the participants would be considered to have had a smartphone addiction as this is above the threshold of what would have been considered one. After the test period, this participant increased their score, further establishing that status. However, all of the participants regarded their own smartphone habits to have been problematic both before and after the test period, regardless of the test score they received. On average, Test Group 1 scored high on Q9: *Using my smartphone longer than I had intended*, with an average score of 5. This further increased after the test period, resulting in a score of 5.2. This correlates to the experiences they reported in the interview sessions, addressing this issue both before and after the test period.

All but one participant from Test Group 1 reduced their screen time from before to after the test period. The participants whose screen time had decreased were all positive towards the timer function of the application (BCT 1 and BCT 12), which might be a contributing factor as to why they spent less time on their smartphone. The participant whose screen time had increased rightfully thought so during the interview session, but also pointed out that they did not attribute this to the application but rather life circumstances. However, this still indicates that the application might not have been effective enough to intervene.

It is important to point out that one participant was unable to share their screen time, making it impossible to know if their screen time had increased or decreased during the test period. However, in their own experience they thought that their smartphone use might have decreased a little bit, but that they still have not had more time over for other tasks, nor that their procrastination or productivity had changed in a positive manner.

5.2.2 G2: Quantitative data

Before the test period four participants in Test Group 2 scored 33 or higher on the SAS-SV, confirming that those four are considered to have a smartphone addiction, even though all six participants considered themselves using their smartphone more than they would like. On the SAS-SV scale Q4: *Will not be able to stand not having a smartphone,*

and Q9: *Using my smartphone longer than I had intended*, both scored high average scores of 5.3 and 5, confirming these being bigger problems collectively among the participants before the test period. Considering all participants use social media platforms these findings can correlate. Social media, as Xu et al. (2026) explains, uses addictive elements such as endless scrolling that makes the user have a hard time self-regulating which can lead to procrastination and more time spent on your smartphone than preferred. Also, getting new content regularly gives you immediate rewards which can be considered as motivation for the users, and one clear explanation for why it is hard for the users to stop and end up using their smartphone longer than planned (Fogg, 2009).

All participants had a good experience and said they reflected more over their smartphone use, while using Application 2. This can also be connected to all participants in Test Group 2 decreasing their screen time clearly on the applications they choose to restrict, which were the ones they thought were the most disturbing. The total average screen time during the test period had four of six participants decreasing their screen time which can be a result of the nudges being effective and making the participants use their smartphone more attentively. The screen time decreased can be linked to the nudges, making the applications less accessible and notification restricted, while also giving the user a prompt asking if it is a good idea to use the app at the time. This shows a change in the participants screen time, which could be explained by a new behaviour where their motivation for using the apps have changed. They don't get prompts from the apps because the applications don't give them notification, and when entering the app they instead get a prompt questioning the user whether it is a good idea for using the application, which makes the user reflect and also makes the application less accessible (Fogg, 2009). Lastly, while the participants reflect they might realize they don't have motivation to open the application, and because of that instead closes it and decides to do something else. Like that the nudges target the prompt, availability and motivation in the participants and makes them change their behaviour regarding time spent on their smartphone (Fogg, 2009).

That all participants didn't get stuck in their smartphone after the test period, compared to before, also shows in the scores of the SAS-SV. The average score of Q9:*Using my smartphone longer than I had intended* decreased to 3.7 (–1.3). Also, Q2: *Having a hard time concentrating in class, while doing assignments, or while working due to smartphone use,* decreased its average score the most after the test period (1.5) with –2 points. This proves that the participants explaining they didn't get as disturbed by their smartphone, and that keeping focus on other tasks got easier, was true. Not getting notifications as a prompt to take up your smartphone can be an explanation for this new behaviour (Fogg, 2009).

5.3 Summary

The research question that this study aimed to answer was "Which behaviour-influencing design elements appear most effective in addressing procrastination related to

problematic smartphone use, and why?". The results and analysis points to that both applications had positive aspects related to them. Test Group 2 had one more participant that experienced improvement in their tendencies towards procrastination than Test Group 1 had, which means that there is not a significant difference in the results connected to procrastination. There were also not a majority within the group, with there being three out of six and two out of five participants that experienced this improvement respectively. However, there were other indicators that pointed towards the effectiveness of the behaviour-influencing design elements within the applications.

The participants from Test Group 1 perceived their smartphone usage as excessive both before and after the test period. The majority of the participants were still unsatisfied with their behaviour towards their smartphone use, leaving only two participants wanting to potentially continue using Application 1. On the other hand, the participants from Test Group 2 experienced that their smartphone usage had become one where quality of consumption was the focus. Although all of the participants experienced their usage as problematic before the test period, they experienced that they had not got as stuck on their smartphones, but rather reflected over what they wanted to spend their time on, showing more satisfaction when the test period was over.

The screen time is harder to compare as Test Group 1 was missing one result as well as having one participant withdraw their participation in the study. However, Test Group 1 showcased three participants with decreased screen time and one that increased, while Test Group 2 had four that decreased and two that increased. The results showed that Test Group 1 had a greater increase of screen time, while Test Group 2 had a greater decrease. Test Group 2 also had four more participants who had used their smartphone less during the test period than Test Group 1. It is possible that this has to do with their experience of more focused and intentional screen time as a result of Application 2. The data from the interviews shows an overall lower satisfaction with the application from Test Group 1. The results from the SAS-SV also showed that their final score was higher than that of Test Group 2, meaning that they on average showed greater tendencies of smartphone addiction. However, the score did not differ significantly between the two groups.

Although the results from the study do not show a conclusive tendency towards improvement of procrastination, it is possible to see indications that Application 2 was more effective at addressing the issue. The participants showed greater satisfaction with the application after using it for a week, stating that their problematic smartphone use had started to become a more intentional one. Their SAS-SV score also dropped by significantly more than Test Group 1 did, showing a great improvement although the average end result between the two groups did not vary significantly. However, the nudge strategies from Olson et al. (2023) seemed to have been both effective and appreciated by the participants. The ten nudge strategy uses motivation, ability and prompt in order to help the user with their smartphone habits. For example, all six participants from Test Group 2 reported that the notification that pops up while opening a limited application prompted them to reconsider whether or not they really wanted to spend their time on it.

This nudge seemed to have helped several participants feel more focused and intentional, reducing aimless screen time and also somewhat procrastination.

6 Discussion

This chapter includes a discussion of the findings from the study, as well as a discussion about the contribution of the results to the interaction design knowledge. The methods that were used are also discussed, as well as the limitations of the study. The research question is also being answered, making the study reach its conclusion.

6.1 Discussion of Findings

The research question guiding this study was “Which behaviour-influencing design elements appear most effective in addressing procrastination related to problematic smartphone use, and why?”.

In the end, the results showed a tendency of Application 2, where nudges were the main design element, as being more effective for the participants in this study. Although the results were not conclusive, it is still indicative of this being the case. Several participants stated a higher satisfaction with their screen time, as well as better focus and productivity. Olson et al. (2023) writes about nudges as strategies that can target a user's motivation and ability, prompting them to make better choices for themselves regarding their smartphone habits. This seems to have been the case for the participants of Test Group 2, as the nudges included in the application helped them make more informed decisions about their smartphone use. For example, both notifications and accessibility seemed to have been effective with providing a more intentional smartphone use.

The application that focused more on BCTs did not perform as well in this study. The reason for this might be that BCT-based interventions may require a longer period of engagement before behavioural changes become noticeable (Michie et al., 2015), which is one potential explanation of the results. However, several participants of Test Group 1 stated that it was hard for them to find the motivation to use the application, as well as to remember to use it in the first place. This is problematic because the application depends on repeated and intentional use. If users do not integrate it into their everyday routines, its behaviour-changing potential becomes limited. The gamification aspects of the application did not seem to be motivation enough in order to combat the procrastination in the participants. Pereira et al. (2025) describes gamification as a potentially useful tool to encourage positive behaviour and increase the motivation in users, as motivation has to be one of the main targets to combat procrastination. However, this did not seem to be the case, at least not in this particular instance as it was explicitly stated by a few participants that they did not feel motivated by the features of the application.

In the study, the applications that were chosen did not contain all of the ten nudges (Olson et al., 2023), and only tested one of the BCT-clusters from Kirchner-Krath et al. (2025). This, as well as the smaller test groups and limited test period, is of course important to keep in mind while reviewing the results from this study. Had a different combination of design elements been included, or if the applications had been tested for a longer time by

more participants, the results might have looked different. However, what the results suggest is that nudges seem to be the more effective alternative in order to receive instant, positive behaviour changes related to problematic smartphone use, as well as indications and habits that might improve procrastination.

6.2 Contribution to Interaction Design Knowledge

This study contributes to the field of Interaction Design by approaching the interconnected problem of procrastination and problematic smartphone use with behavioural-influencing design. A problem area that, to the authors' knowledge, has not been previously studied.

The findings contribute to knowledge of indications that nudge strategies, more specifically Notifications, Accessibility and Social media (Olson et al., 2023), appeared effective in reducing problematic smartphone use, and to some extent, procrastination. These results support previous studies of nudges being an effective behaviour-influencing method against problematic smartphone behaviour (Brockmeier et al., 2025; Liu et al., 2025; Olson et al., 2023), while also extending this understanding through this study with new findings indicating it may also influence procrastination.

From a design perspective, this study contributes to practical knowledge of how behavioural-influencing design elements can be implemented in design solutions. The results indicate that nudges that target the users during moments of habitual use, like when they open up an application, or get stuck doomscrolling without purpose, can help the user to reflect and rethink their behaviour.

The finding of this study indicates that behavioural-influencing design can contribute to more intentional and meaningful smartphone use, which can support better focus and productivity. These findings may help interaction designers developing digital solutions for habitual or problematic smartphone behaviours in other contexts.

6.3 Discussion of Method

Looking back at the methods used in this study, some worked well while others could have been done differently to get an even more validated and reliable result. To begin, a convenience sampling was used to find suitable participants for our study . This method can contribute to the whole target group not being represented, but only the ones invited to participate (Jacobsen, 2017). This limits the generalisability of the findings. Also, the only requirements for participation were to be an adult (+18), and to use your smartphone more than preferred. This means the study was not taking into account demographic factors like gender, specific age, occupations, or others. This means that the two test groups could have very different participants, with different demographic backgrounds and representation of the target group, which could affect the outcome of the study as the groups are not directly comparable (Jacobsen, 2017).

Considering the A/B test conducted in the study, two completely different applications were chosen to be compared. The applications included different combined behaviour-influencing elements, where one incorporated BCTs and one different nudge strategies. Comparing two completely different applications, and behaviour-influencing frameworks, made it difficult to understand what really contributed to the outcome of the test, as A/B test primarily supports comparisons of outcomes between variants, rather than the explanations of the underlying causes (Quin et al., 2024) The result now only gives an indication of which application and design elements appeared most effective overall. To get a more detailed understanding of why the elements were more effective, more in-depth studies on one or a few combined behaviour-influencing elements should be considered.

The quantitative methods used, SAS-SV and screen time statistics, were helpful to get more data regarding behaviour of the participants' smartphone usage, before and after the test period. In retrospect, it would have been helpful to have added a measurement for procrastination, to get a clearer understanding of how it compared to before and after the test period and to validate the results of the test period related to procrastination (Jacobsen, 2017). However, the built-in screen time statistics of the users smartphone was not as flexible as desired. For once, some participants could only show weekly statistics, where the statistics from the latest week started on Monday, and not from the current day. This meant that if the participant shared a statistic on another day than Monday the statistic only showed the last days since the last Monday, and not the total last seven days. This meant that the test either had to start on a Monday, or the participants had to share each individual day of the last week, and then the authors had to manually calculate the total. Also, one of the participants did not have the screen time statistic function on their smartphone and therefore could not share this data. This was a limitation, as the data collected from the test group were not fully comparable (Jacobesen, 2017).

The semi-structured interviews were a good complement to the other methods to gain a deeper understanding of the participants' experience during the test period. To be considered, it could be difficult for the interviewer to not be biased when asking questions. Also, the flexible nature of the method makes it hard to ensure the interviews are equal for all participants (Jacobsen, 2017).

Lastly, the thematic analysis was used to analyze the qualitative data. This method was helpful for finding patterns in the collected data, however it has some limitations. When coding, establishing themes, and finding patterns, it is all done with the subjective interpretations of the authors. Nuance can be lost when creating codes and forming themes, and the authors can have a bias affecting the creations of themes (Jacobsen, 2017).

6.4 Limitations

This study has a couple of limitations that have been identified. First, the study had a small sample size, initially of six participants in each test group which later dropped to

five in one group. This is a limitation since the generalisability of the results is being reduced (Jacobsen, 2017). The findings are still an indication of what behaviour-influencing design elements may be effective against procrastination caused by problematic smartphone use.

Second, the length of seven days for the test period can be considered short. With only seven days it can be hard to see a real change in behaviour when using BCTs, as many interventions are designed to support gradual progress over time, implying that behavioural change may require longer than one week (Michie et al., 2015). For example, participants may have other factors that will make them decrease or increase their screen time during the week that might not be linked to the applications tested. Therefore, testing for a longer period of time would give better indications for how the behaviour changes over time, and if it would last.

In addition, the study only tested a limited combination of BCTs against a limited combination of nudges. How these are combined, and which of the elements included in the applications tested, affects the results. This study only tested one set of combined BCTs against one set of combined nudges. There are still plenty more nudges and BCTs that can be combined in different ways that can be effective against procrastination caused by problematic smartphone use which limits the generalization of this study (Jacobsen, 2017). However, this study presents indications of what works on this target group. Future studies will have to test more combinations of behaviour-influencing elements to see if it can provide a more effective result.

7 Conclusion

This chapter presents the conclusion of the study. It answers the research questions, discusses societal and ethical implications, and outlines directions for future studies.

The final conclusion of this study is that Nudges seem to be the most effective behaviour-influencing design element in addressing procrastination related to problematic smartphone use, at least in the short term and for immediate results. Although only half of the participants of Test Group 2 reported lowered procrastination rates, they reported implications of a less interruption from their smartphone leading to better focus. The findings of the study also concluded that nudges had an effect in the users smartphone usage decreasing unintentional and habitual usage to increase more reflective and purposive use. The findings also concluded that Application 1 with BCT-based elements required stronger user motivation to get user engagement.

Our study primarily benefits the users of smartphones who unintentionally use their smartphone instead of doing other tasks or activities they intend to. To be considered, it can be problematic to intentionally interfere to change someone's habitual behaviour. However, behavioural-influceing elements are already used today as a marketing strategy to get users hooked, and are proven to work. Our study provides design solutions to work against the already existing strategies, helping the user to use their smartphone on their own terms. To not force the behaviour change this study included methods that allows the user to restrict applications themself, according to their own goals, or when they want to focus.

Something to take into account is that all smartphone usage isn’t negative for the user and by restricting applications you also restrict the positive benefits like social communication. This can be seen as a potential consequence of this study's proposed solution. However, the interventions never totally constrained the user, just made it harder to access applications. This makes it possible for the user to always use their smartphone, but they will be nudged into reflecting if it is necessary.

For future research on this topic, it would be of interest to explore what effect the different design interventions would have if the time horizon would not be as limited as it was in this study. If the applications were tested over a period of a few weeks or longer, the results might paint a picture of the sustainability of the existing design solutions, as well as potential improvements. It would also be beneficial to include more participants in a future study, in order to gain an even broader perspective on the results and further input about the effectiveness of different design approaches.

Another suggestion for future research is to test more or different combinations of BCTs and nudges, or even combine both of them in order to gain a clearer understanding of what would be most beneficial in order to address procrastination and problematic smartphone use.

References

Agarwal, N. K., & Lu, W. (2022). The Yin and Yang of smartphones: an interview study of smartphone use and its effects. *Global Knowledge, Memory and Communication*, 71(6/7), 432–448. https://doi.org/10.1108/GKMC-01-2021-0018

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative research in psychology*, 3(2), 77-101. https://doi.org/10.1191/1478088706qp063oa

Brockmeier, L. C., Mertens, L., Roitzheim, C., Radtke, T., Dingler, T., & Keller, J. (2025). Effects of an intervention targeting social media app use on well-being outcomes: A randomized controlled trial. *Applied Psychology: Health and Well-Being*, 17(1), e12646. https://doi.org/10.1111/aphw.12646

Ertemel, A. V., & Ari, E. (2020). A Marketing Approach to a Psychological Problem: Problematic Smartphone Use on Adolescents. *International Journal of Environmental Research and Public Health*, 17(7), 2471. https://doi.org/10.3390/ijerph17072471

Fogg, B. (2009). A behavior model for persuasive design. *Proceedings of the 4th International Conference on Persuasive Technology*, 1–7. https://doi.org/10.1145/1541948.1541999

Geng, J., Han, L., Gao, F., Jou, M., & Huang, C.-C. (2018). Internet addiction and procrastination among Chinese young adults: A moderated mediation model. *Computers in Human Behavior*, 84, 320–333. https://doi.org/10.1016/j.chb.2018.03.013

Jacobsen, D. I. (2017). *Hur genomför man undersökningar? : introduktion till samhällsvetenskapliga metoder* (Andra upplagan). Studentlitteratur.

Johansson, L.-G. (2018). *Introduktion till vetenskapsteorin* (Fjärde upplagan). Thales.

Kirchner-Krath, J., Schmidt-Kraepelin, M., Schmähl, K., Schütz, C., Morschheuser, B., & Sunyaev, A. (2025). Behavior Change Support Systems for Self-Treating Procrastination: Systematic Search in App Stores and Analysis of Motivational Design Archetypes. *Journal of Medical Internet Research*, 27(1), e65214. https://doi.org/10.2196/65214

Kwon, M., Kim, D.-J., Cho, H., & Yang, S. (2013). The Smartphone Addiction Scale: Development and Validation of a Short Version for Adolescents. *PloS One*, 8(12), e83558. https://doi.org/10.1371/journal.pone.0083558

Li, L., Gao, H., Xu, Y. (2020). The mediating and buffering effect of academic self-efficacy on the relationship between smartphone addiction and academic procrastination. *Computers & Education*, 159, 104001. https://doi.org/10.1016/j.compedu.2020.104001

Liu, Z., Chen, Y., Bian, T., Xiang, Y., Yuan, R., & Yang, H. (2025). Associations among smartphone addiction, chronotype, and bedtime procrastination: bayesian network analysis and pilot intervention. *Current Psychology (New Brunswick, N.J.)*, 44(5), 3326–3339. https://doi.org/10.1007/s12144-025-07407-9

Lukas, C. A., & Berking, M. (2018). Reducing procrastination using a smartphone-based treatment program: A randomized controlled pilot study. *Internet Interventions*, 12, 83-90. http://dx.doi.org/10.1016/j.invent.2017.07.002

Michie, S., Wood, C. E., Johnston, M., Abraham, C., Francis, J. J., & Hardeman, W. (2015). Behaviour change techniques: the development and evaluation of a taxonomic method for reporting and describing behaviour change interventions (a suite of five studies involving consensus methods, randomised controlled trials and analysis of qualitative data). *Health Technology Assessment (Winchester, England)*, 19(99), 1–188. https://doi.org/10.3310/hta19990

Nyberg, R., Tidström, A. (2012). *Skriv vetenskapliga uppsatser, examensarbeten och avhandlingar* (Upplaga 2). Studentlitteratur.

Olson, J. A., Sandra, D. A., Chmoulevitch, D., Raz, A., & Veissière, S. P. L. (2023). A Nudge-Based Intervention to Reduce Problematic Smartphone Use: Randomised Controlled Trial. *International Journal of Mental Health and Addiction,* 21(6), 3842–3864. https://doi.org/10.1007/s11469-022-00826-w

Pereira, M. B., de Classe, T. M., & Siqueira, S. W. M. (2025). Investigating gamification to reduce procrastination: Systematic literature review. *Journal in Interactive Systems*, 16(1). https://doi.org/10.5753/jis.2025.5412

Quin, F., Weyns, D., Galster, M., & Silva, C. C. (2024). A/B testing: A systematic literature review. *The Journal of Systems and Software*, 211, Article 112011. https://doi.org/10.1016/j.jss.2024.112011

Xu, S., Pang, H., & Huang, D. (2026). Problematic TikTok use and subjective well-being of higher vocational students: Significant roles of academic procrastination, emotional dysregulation, and social avoidance. *International Journal of Mental Health and Addiction*. https://doi.org/10.1007/s11469-025-01635-7

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Appendix 1. Search String

Date: 2026-02-25

Database: Scopus, OneSearch

Search string:

("persuasive technology" OR "behavior change support systems" OR "behavior change techniques" OR "motivational design" OR nudge OR gamification OR gamify OR "uses and gratifications theory") AND ("mobile apps" OR "mobile health" OR "mobile phone" OR "interactive systems" OR smartphone) AND (procrastination OR "academic procrastination" OR Postpone OR "internet addiction" OR "smartphone addiction" OR "mobile addiction" OR "problematic mobile use")

Limits: Articles, English, Peer-reviewed

Result: 18 documents found

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Appendix 2. Informed Consent Form

Samtyckesblankett för att delta i Beteendepåverkande design mot prokrastinering till följd av problematiskt mobilanvändande

Syftet med studien är att undersöka vilka beteendepåverkande designtekniker som är effektivast mot prokrastinering hos människor som upplever ett problematiskt mobilanvändande. Vi kommer att undersöka detta med en experimentell metod där två testgrupper kommer att få använda var sin app med beteendepåverkande designelement under en 7-dagarsperiod. Varje deltagare kommer att intervjuas om sin upplevelse före och efter testet, genomföra ett skärmberoendetest, och dela sin skärmtid. Intervjun kommer att spelas in och därefter transkriberas.

Genom att skriva under den här samtyckesblanketten godkänner du att dina personuppgifter behandlas inom ramen för uppsatsen/studien som beskrivs ovan. Du kan när som helst dra tillbaka ditt samtycke genom att kontakta någon av kontaktpersonerna nedan. Dina personuppgifter kommer därefter inte längre att bevaras eller behandlas vidare utan annan laglig grund.

Den data som kommer samlas in från dig är ljudinspelade intervjusvar, svar från skärberoendetest samt data över din skärmtid. Dina personuppgifter kommer att behandlas från mars 2026, till mars 2028. De data som samlas in under denna studie kommer endast att vara tillgängliga för den student som genomför studien, handledaren och examinatorn, och kommer enbart att användas för att slutföra uppsatsen. All insamlad data kommer att raderas när uppsatsen formellt har godkänts och examinationsprocessen är avslutad.

Du har alltid möjlighet att få information om vad som registrerats om dig eller ha synpunkter på behandlingen eller de uppgifter som samlats in genom att kontakta någon av kontaktpersonerna nedan eller lärosätets dataskyddsombud på dataskyddsombud@lnu.se. Klagomål som inte kan lösas med Linnéuniversitetet kan lämnas till Integritetsskyddsmyndigheten.

…… ……
Underskrift Ort och datum

……
Namnförtydligande

Kontaktuppgifter:

Studentens namn: Linnea Karlsson
Studentens e-postadress: lk223sh@student.lnu.se

Studentens namn: Tea Sjöblom Engvall
Studentens e-postadress: ts223fj@student.lnu.se

Handledarens namn: Juan P. Velásquez
Handledarens e-postadress: juan.velasquez@lnu.se

Examinatorns namn: Päivi Jokela
Examinatorns e-postadress: paivi.jokela@lnu.se

Appendix 3. Questions before the test period

1. Tycker du själv att du lägger mer tid på din mobil än vad du vill?
2. Vad hade du velat lägga tiden på istället för på mobilen?
3. Upplever du, på grund av ditt mobilanvändande, skjuter upp uppgifter i din vardag?
4. Vad gör du oftast när du använder din mobil?

Appendix 4. SAS-SV

1. 1. Jag missar planerat arbete på grund av smartphone-användning.
2. 2. Jag har svårt att koncentrera mig på lektioner, när jag gör uppgifter eller när jag arbetar på grund av smartphoneanvändning.
3. 3. Jag känner smärta i handlederna eller i bakre delen av nacken när jag använder en smartphone.
4. 4. Jag skulle inte klara av att vara utan en smartphone.
5. 5. Jag känner mig otålig och rastlös när jag inte håller i min smartphone.
6. 6. Jag tänker på min smartphone även när jag inte använder den.
7. 7. Jag kommer aldrig att sluta använda min smartphone även om mitt dagliga liv redan påverkas mycket av den.
8. 8. Jag kontrollerar ständigt min smartphone för att inte missa konversationer mellan andra personer på Twitter eller Facebook.
9. 9. Jag använder min smartphone längre än jag hade tänkt.
10. 10. Personerna runt omkring mig säger att jag använder min smartphone för mycket.

Appendix 5. Interview questions

1. Upplever du själv att du lagt mer tid på din mobil än vad du velat senaste veckan?
2. Har du haft tid över till annat? Vad har du gjort istället?
3. Hur har din upplevelse varit av att använda appen?
4. Har du märkt någon skillnad i din vardag?
5. Vilka positiva effekter upplever du att användande haft?
6. Vilka negativa effekter upplever du att användande haft?
7. Kan du beskriva vad du tycker är det bästa med appen?
8. Kan du beskriva vad som inte fungerade i appen?
9. Kommer du att fortsätta användandet av appen?
10. Upplever du att du har använt mobilen mindre? Om ja, vid vilka tillfällen? Och vad har du gjort med tiden istället?
11. Upplever du att du har blivit produktivare? Om ja, på vilket sätt?
12. Upplever du att du prokrastinerat mindre då du använt dig av appen?

Appendix 6. Thematic Analysis Themes

Testgrupp	Huvudtema	Undertema
1	Smartphoneanvändning	Prokrastinering grundat i mobilanvändning
		Mobilanvändning innan
		Användningsområde telefon
		Vill istället lägga tid på
1	Effekter av appanvändning	Inga positiva effekter av appen
		Osäker effekt av appen
		Positiva effekter av appen
		Indirekta effekter
1	Upplevelse av appanvändning	Upplevelse användning av appen
		Upplevelse funktioner i appen
		Inte använt appen så mycket
		Ej fortsatt användning
		Fortsatt användning
2	Mobilanvändning	Mobilanvändning innan
		Önskad aktivitet istället för mobil
		Prokrastinering innan
2	Upplevelsen av app	Tankeställare
		Notiser
		Positiv upplevelse
		Sämre med applikationen
		Flyttat användande
2	Effekt av appen	Mobilanvändning Efter
		Mer tänkt skärmtid
		Minskat onödigt scrollande
		Prokrastinering efter
		Produktivitet

Appendix 7. Användning av AI

I denna studie så har AI använts vid två tillfällen. Först användes AI-verktyget Scopus AI (Elsevier, 2026) för att bidra med relevanta söksträngar och sökord inom vårt forskningsområde. Vi använder verktyget som ett komplement till våra egna sökord, där resultatet från Scopus AI utvärderades utifrån relevans innan passande sökord adderades till vår redan befintliga söksträng.

Det andra tillfället där AI användes var vid översättning av forskningsplanen. Vår forskningsplan skrevs på svenska, medan den slutliga rapporten är skriven på engelska. För att kunna återanvända redan skriven text från forskningsplanen använde vi oss av AI-verktyget Deep L Translator (Deep L, 2026). AI-verktyget översatte texten från svenska till engelska. Efter översättningen har texten bearbetats för att säkerställa att texten behållit samma budskap.

Elsevier, B. V. (13 maj 2026). *Scopus – Homepage.* Scopus. https://www-scopus-com.proxy.lnu.se/pages/home#scopus-ai

Deep L, (13 maj 2026). *Deep L Translator: The world's most accurate translator. Deep L.* https://www.deepl.com/en/translator

Appendix 8. Arbetsfördelning mellan skrivpartners

Arbetsfördelningen i denna rapport har fördelats lika mellan skrivpartners. Båda har deltagit och genomfört alla moment i studien så som studerande av tidigare forskning, utforma forskningsfråga, planering av genomförande, rekrytering av deltagare, modererat intervjuer och undersökningar, transkribering, tematisering, analys och skriva rapport.